

Lumbini ICT Campus, Nawalpur Gaindakot

Smart BMI & Body Fat Analysis System

A web-based health monitoring platform with personalized diet and exercise recommendations

Rupesh Phyauli Magar | Ujjwal Acharya |

Submitted to: Modnath Acharya

Health awareness is rising, but digital tools are lagging behind

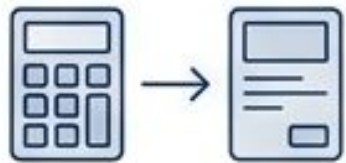
The Status Gap

Many individuals remain unaware of their precise BMI and body fat percentage status.



The Analysis Gap

Most existing calculators provide only basic classification results without deep health analysis.



User
Intent



Tool
Capabilities

24.5

The Guidance Gap

Users lack an easy-to-use platform that links their body condition to personalized diet and exercise guidance.



Beyond Basic BMI



Basic Classification

Overlooks body composition. Stops at basic weight-to-height ratios.



Deeper Health Analysis

The true fitness indicator. Accounts for age, gender, and actual body composition.

Most online tools stop at BMI. We go further.

The Current Gap in Digital Health



Blind Spots

Users do not know their true body fat status.



Generic Advice

Existing platforms offer one-size-fits-all suggestions.



No Guidance

Lack of integrated, easy-to-use action plans based on specific body conditions.

A blueprint for intelligent health monitoring



1. Accurate Calculation

Developing a precise engine to calculate both Body Mass Index and Body Fat Percentage.



2. Tailored Diet Plans

Providing personalized nutritional suggestions matched specifically to the user's health condition.



3. Targeted Exercise

Recommending suitable physical activity plans based dynamically on the calculation results.

Driving recommendations through rule-based logic

The Data Foundation: Healthcare dataset insights derived from NHANES (National Health and Nutrition Examination Survey).

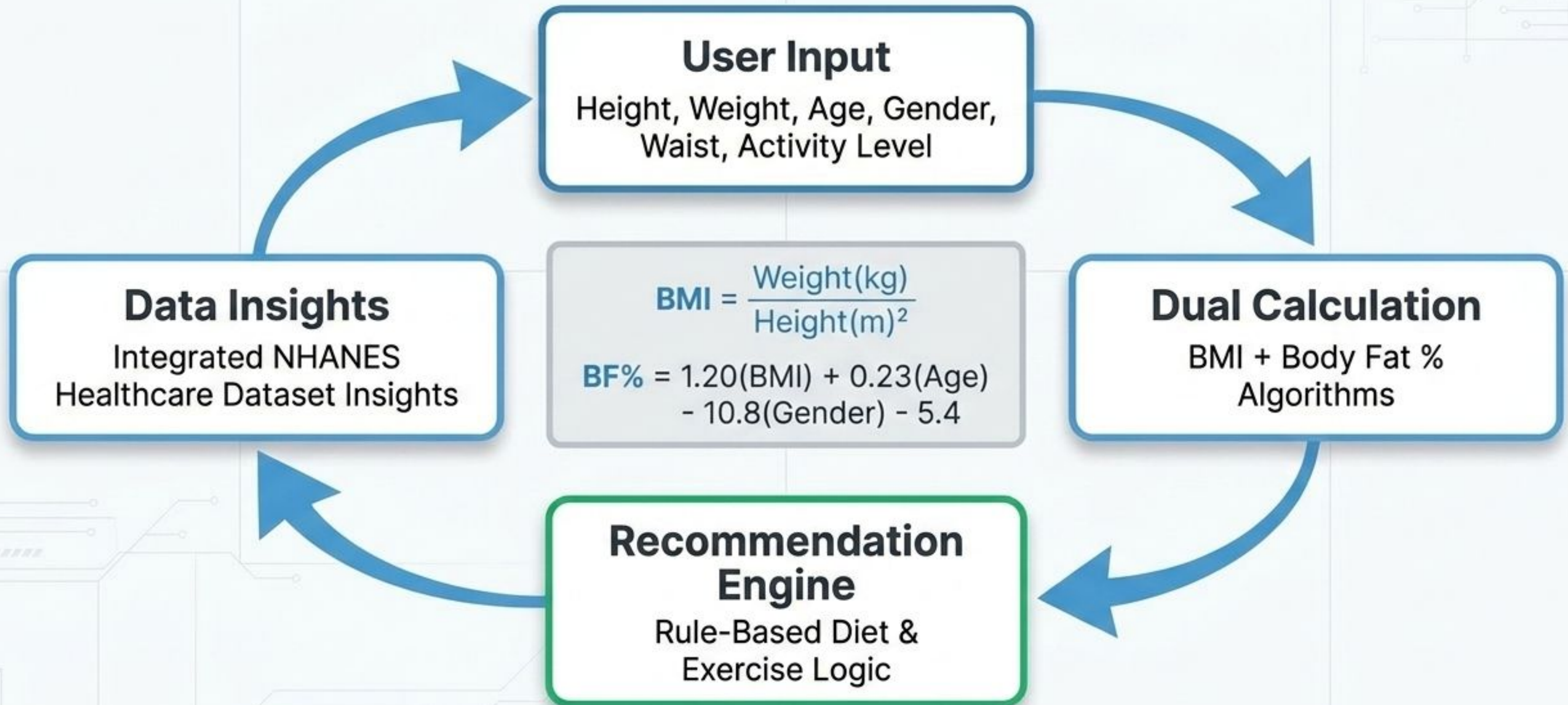
The Core mechanism:
A Rule-Based
Recommendation
System.

If [Category = High Body Fat]
+ [Activity = Low]

Then Route to [Caloric Deficit Diet Plan]
+ [Cardio-Focused Exercise Plan].

This integration creates a system that acts intelligently upon the calculated health metrics rather than just displaying them.

Core System Features



Expected Outcomes: A complete health monitoring system



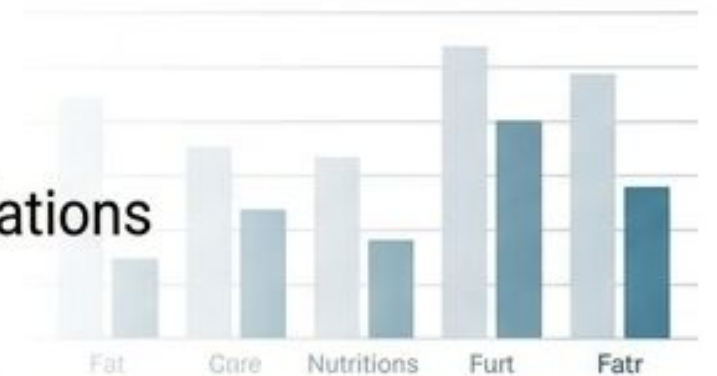
Frictionless Calculation

Users will be able to instantly calculate their BMI and estimate body fat percentage without specialized equipment.



Intelligent Nutrition

The system will automatically generate personalized diet recommendations tailored to individual health statuses.



Actionable Fitness

Users will receive specific, categorized exercise plans perfectly suited to their current body condition and fitness capacity.



Foundation & References

World Health Organization (WHO)

– Obesity and BMI standards.

Centers for Disease Control (CDC)

– Adult BMI classifications.

American Council on Exercise

– Body Fat Percentage parameters.

NHANES

– Health and Nutrition Examination Survey datasets.

Django & Python

– Core backend official documentation.

Pandas & NumPy

– Data analysis and processing documentation.

Scikit-learn

– Advanced data modeling references.

Jupyter Notebook

– Data exploration documentation.

Thank You!

Smart BMI & Body Fat Analysis System

We are happy to answer your questions

Rupesh Phyauli Magar · Ujjwal Acharya ·

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